2.8 Circular Motion

Learning Goals (Students will be able too….)

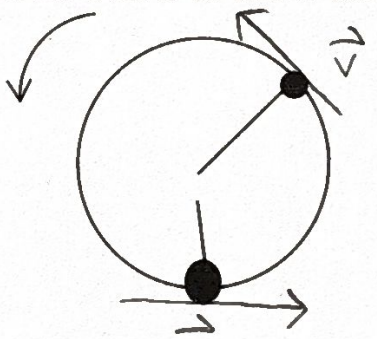
- Define centripetal force
- Learn about the direction of centripetal force
- Identify the forces that contribute to centripetal force
- Calculate the centripetal acceleration of an object

Accelerating Frames of References

1) Uniform circular motion is exactly what it sounds like – something moving in a circle at a constant speed.
   a) Is an object in uniform circular motion accelerating? Explain.

      Yes, becawse it is changing direction, and acceleration is the rate of change in velocity, which accumuls, for both speed and direction.

   b) Draw velocity vectors on the circle below to show the direction of the velocity at the points indicated.



   c) The velocity is always tangential to the path and perpendicular to the radius of the circle.

Demonstration

https://www.youtube.com/watch?v=ID0R43My4Co

Discussion as a Class:

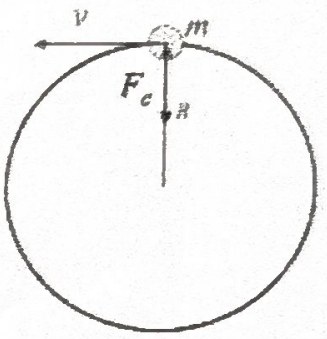
1) When does he need to lift the hoop in order to knock the cards down?
2) In what direction is the net force? What does it say about the direction of acceleration?

When the instantaneons velocity is pointing towards the cards, the net force is perpendicular to the velocity, so the acceleriation is towards the center of the hoop.

This centre-seeking force is called the **centripetal force**. The centripetal force is not a force in and of itself. Rather, it is the **net force** in the direction of acceleration and is made up of other forces.

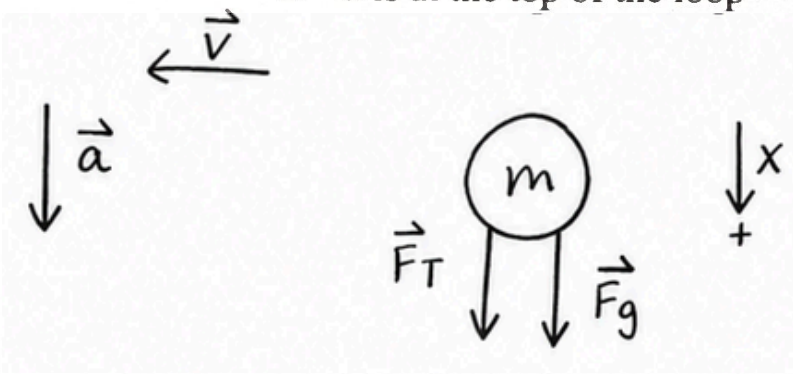
Analyzing the Forces at Play

For each of the following draw a **Free Body Diagram** and state the force or forces that contribute to the centripetal force. Remember that for centripetal acceleration to occur, the **net force must be towards the centre of the circle**.



1) A ball is swinging on a string in a vertical loop. Consider the ball when

   a. It is at the top of the loop



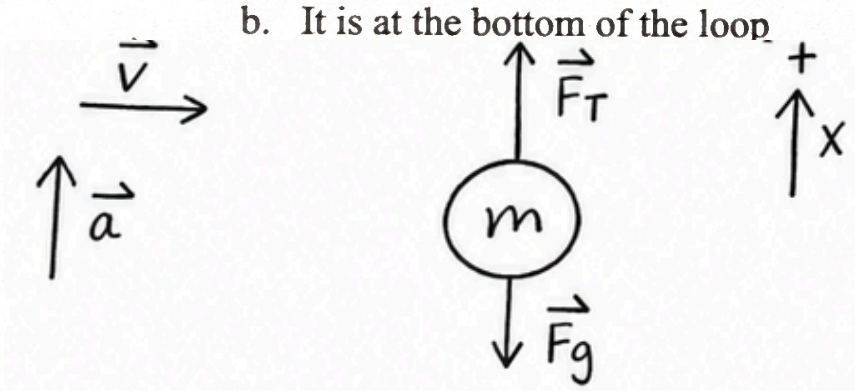
centripetal force

$\vec{F}_{net} = \vec{F}_c$

$= \vec{F}_g + \vec{F}_T$

$F_c = F_g + F_T$

   b. It is at the bottom of the loop



$F_{net} = F_c$

$F_{net} = F_T - F_g$

$F_c = F_T - F_g$

   c. If the net centripetal force is to remain constant, what does that tell you about the tension in the string at the two different places?

At the top: $F_{T\,top} = F_c - F_g$

At the bottom: $F_{T\,bottom} = F_c + F_g$

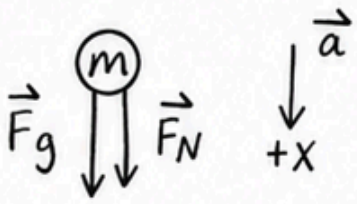
$\therefore F_{T\,bottom} = F_{T\,top} + 2F_g$

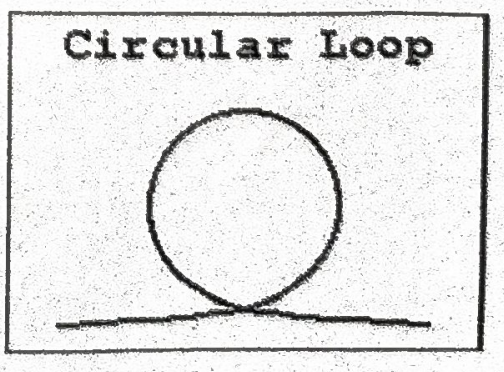
$F_{T\,bottom} = F_{T\,top} + 2mg$

$\therefore$ The tension is greater at the bottom by $2mg$

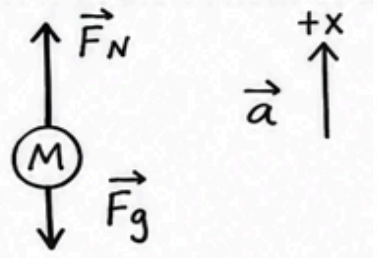
2) A roller coaster doing a loop the loop. Consider the train when

   a. It is at the top of the loop





b. It is at the bottom of the loop



c. At which location would a person feel heavier? Explain.

A person would feel heavier at the bottom because the normal force is greater at the bottom, so their apparent mass would be greater.

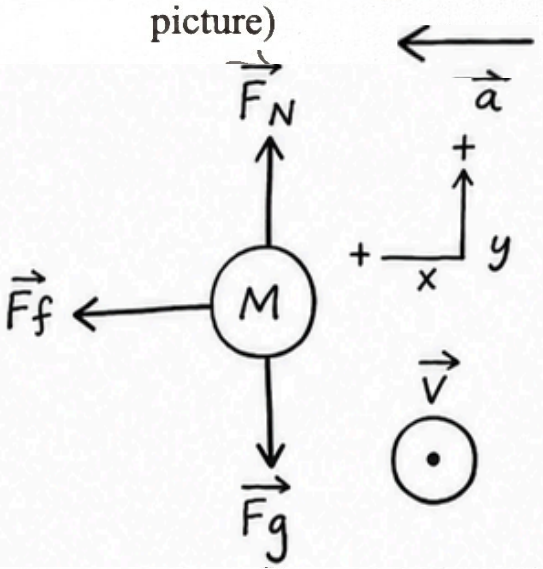
top: $F_{net} = F_N + F_g$ → $F_N = F_{net} - F_g$

F_N: top < bottom

bottom: $F_{net} = F_N - F_g$ → $F_N = F_N + F_g$

$m_{app} = \frac{F_N}{g}$

3) A car is turning on a level road. (coming toward you in the picture)



$F_{net\,y} = F_N - F_g = 0$

$F_N = F_g$

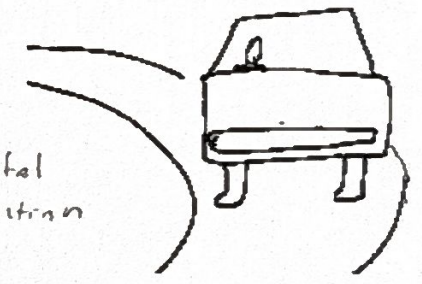
$F_{net\,x} = F_f = ma_c$ (centripetal acceleration)

centripetal acceleration formula:

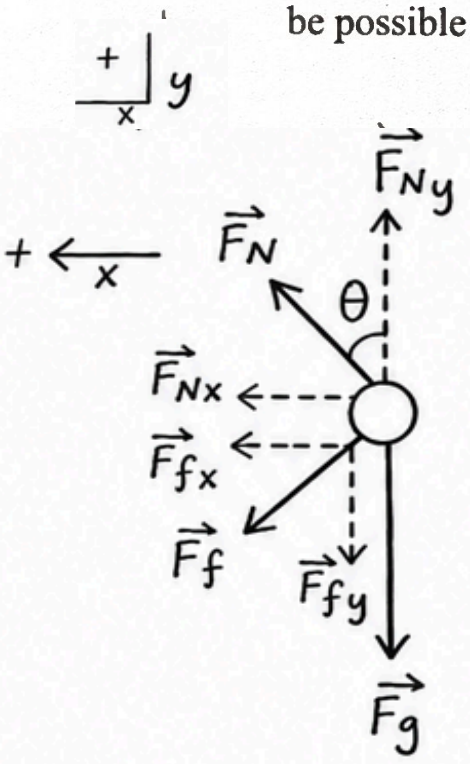
$a_c = \frac{v^2}{r}$

$F_f = ma_c$ $\quad F_f = m \times \frac{v^2}{r}$

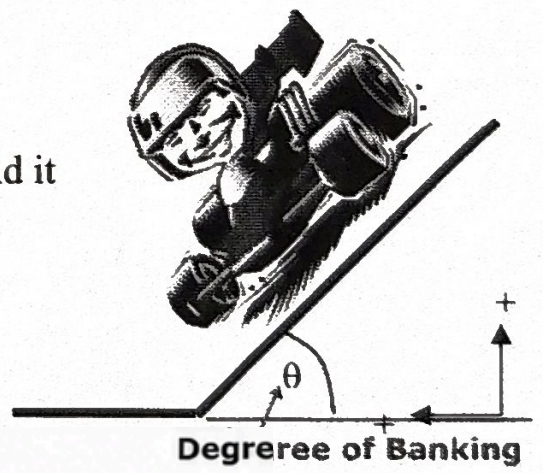
$F_f = \frac{mv^2}{r}$



4) A car is turning on a **banked** road. (coming toward you in the picture). Draw the components of the appropriate force(s) according to the co-ordinate system given in the picture. Would it be possible to corner safely without static friction? Explain.

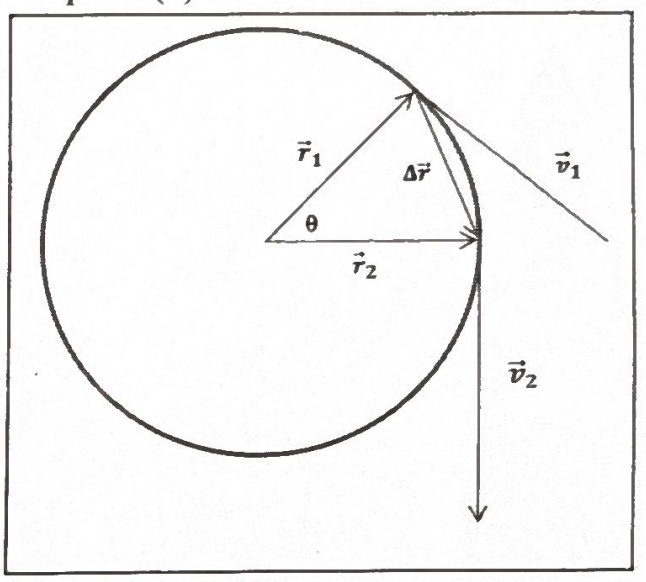


Yes, it is possible to corner safely without static friction at the appropriate speed because the x-component of the normal force provides the centripetal acceleration.



Calculating Centripetal Acceleration

Goal: To derive an equation that will allow us to calculate the magnitude of the instantaneous centripetal acceleration in terms of easily measurable quantities: speed (v) and radius of curvature (r)

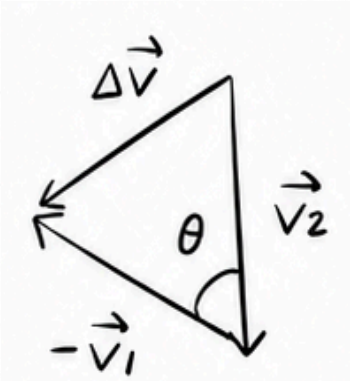


Basic Idea:

We will start with the regular equation for acceleration, $\vec{a} = \frac{\Delta\vec{v}}{\Delta t}$, and just consider a very small time interval.

Derivation:

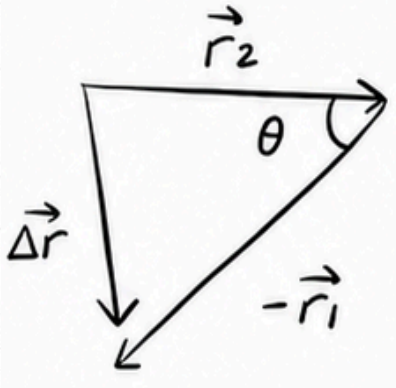
a) Consider an object moving from position $\vec{r}_1$ to position $\vec{r}_2$ as shown in the diagram. Start with the basic definition of acceleration above. We know that when vectors are not co-linear, we have to draw a vector diagram to determine $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$. Do this below. Label it:



$$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$$

$$\Delta\vec{v} = \vec{v}_2 + (-\vec{v}_1)$$

b) Now we need to get a connection between v and r, so draw another vector diagram to show $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$. Label it clearly.



$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\Delta\vec{r} = \vec{r}_2 + (-\vec{r}_1)$$

A couple of simplifications: the magnitude of velocity never changes and neither does the magnitude of the radius. Hence, we can drop our subscripts *i* and *f* ($v_i = v_f = v$ and $r_i = r_f = r$)

c) If the <u>triangles are similar</u> then finish the ratio: $\frac{\Delta v}{v} = \frac{\Delta r}{r}$

d) Now isolate □v in the ratio above:

$$\Delta v = \frac{v\Delta r}{r}$$

e) Take your expression from (d) and sub it into the basic acceleration equation:

$$\vec{a}_{av} = \frac{\vec{\Delta v}}{\Delta t} \qquad a_{av} = \frac{v\Delta r}{\Delta t\, r}$$

f) Notice that $\vec{r}$ is the position vector for the object, so $\Delta\vec{r}$ is <u>change in position</u> which makes $\frac{\Delta\vec{r}}{\Delta t}$ equal to speed which is *v*. Make this substitution in your current acceleration equation and simplify. Check your final result with the textbook. (p 116)

$a = \frac{v}{r}\left(\frac{\Delta r}{\Delta t}\right)$ → displacement magnitude; → time

Over a very short interval of time, the magnitude of the displacement (Δr) becames almost equal to the distance traveled oround the circle.

$\frac{\text{distance}}{\Delta t} = v \quad \therefore \frac{\Delta r}{\Delta t} \approx v \quad \frac{\Delta r}{\Delta t} \to v$ as $\Delta t \to 0$

$$a = \frac{v}{r}(v)$$

$$\boxed{a = \frac{v^2}{r}}$$

Alternative Formulas

Sometimes you know the frequency or period of an object instead of its speed. We therefore need equations for centripetal acceleration that include frequency or period. We will derive these:

Consider an object moves at a constant speed v around a circle of radius r in a time T (called the period since circular motion is periodic).

a) You know $\boldsymbol{v} = \frac{\Delta \boldsymbol{d}}{\Delta \boldsymbol{t}}$. For an object moving in a circle of radius r, what is the distance travelled? What is the time taken for one revolution? Write the formula for the speed of the object in terms of r and T.

One full rotation $= 2\pi r$

time $= T$

$$v = \frac{\text{distance}}{\text{time}} \qquad v = \frac{2\pi r}{T}$$

b) Sub your formula into the centripetal acceleration formula and simplify. Compare your result with your textbook, page 116 bottom.

$$a_c = \frac{v^2}{r} \qquad a_c = \frac{\left(\frac{2\pi r}{T}\right)^2}{r} \qquad a_c = \frac{4\pi^2 r^2}{T^2 r}$$

$$a_c = \frac{4\pi^2 r}{T^2}$$

c) Frequency is the reciprocal of period. Modify your formula in b) to use frequency instead of period. Check your results.

$$f = \frac{1}{T} \qquad f^2 = \frac{1}{T^2}$$

$$a_c = 4\pi^2 r\left(\frac{1}{T^2}\right)$$

$$a_c = 4\pi^2 r f^2$$

Homework

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